

REACT JS

LECTURE 6

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AGENDA

- Lazy loading
- login & protected route demo
- Custom Hooks
- implementing localization
- Environment variables (.env)
- Build & Deploy
- Client side rendering vs Server side rendering

LAZY LOADING & CODE SPLITTING

what is code splitting ?

- Technique to split JavaScript bundle into smaller chunks
- Only load code when it is needed
- Reduces initial load time
- Improves performance and user experience

syntax:

```
const Home = React.lazy(() => import('./Home'));
```

Component is not loaded initially

Loaded when Home is rendered

Must be wrapped in `<Suspense>`

LAZY LOADING & CODE SPLITTING

Suspense component:

- Displays fallback UI while loading
- Required for lazy-loaded components
- Fallback can be spinner or loading text

Syntax: `<Suspense fallback={<Loading />} >`

when not to use lazy loading ?

- Small components
- Frequently used UI elements
- Critical above-the-fold content
- Overuse can harm UX

PRIVATE ROUTES

What is a Private Route?

- A Private Route restricts access to certain pages only to authenticated users.
- If the user is not logged in, they are redirected to Login.

Syntax:

```
import { Navigate } from "react-router";

function PrivateRoute({ isAuth, children }) {
  return isAuth ? children : <Navigate to="/login" />;
}
```

CUSTOM HOOKS

Custom hooks in React are a way to extract and reuse logic that is common across multiple components. They allow you to encapsulate behavior and state management into reusable functions that follow the rules of hooks.

Why Use Custom Hooks?

- **Reusability:** Extract shared logic into a single place to avoid repetition.
- **Abstraction:** Hide complex logic from the component, making it simpler and more focused on rendering.
- **Separation of Concerns:** Separate different concerns within a component, leading to better organization and easier maintenance.

How to Create and Use Custom Hooks

A custom hook is simply a JavaScript function that starts with the word "use" and can call other hooks within it.

LOCALIZATION

What is Localization (i18n)?

Localization means displaying the app in multiple languages based on user choice(e.g. English  / Arabic ).

Core Steps:

1. Create translation files (`en.json`, `ar.json`)
2. Configure `i18n`
3. Use `useTranslation()` hook
4. Change language dynamically

library Link : [React-i18nnext](#)

ENVIRONMENT VARIABLES

What are Environment Variables?

Environment variables are key-value pairs used to store configuration outside the code (e.g. API URLs, ports, environment mode).

What is a .env File?

A .env file is a local configuration file that defines environment variables without hardcoding them into the source code.

example :

```
VITE_APP_API_URL=https://api.example.com
```


ENVIRONMENT VARIABLES

Why Use .env?

-  Improves security (no hardcoded secrets)
-  Easy configuration per environment
-  Same code, different environments (dev / prod)

CSR VS SSR

Client-Side Rendering (CSR)

- The browser downloads JS bundle and builds the page.
- Initial HTML is almost empty; content rendered by JS.
- Fast interactions after page load.
- Example: React SPA (Single Page Application)

CSR VS SSR

Pros:

- Smooth transitions & SPA behavior
- Rich interactivity

Cons:

- Slower initial load
- Poor SEO if not hydrated
- Blank page until JS loads

CSR VS SSR

Server-Side Rendering (SSR)

- The server generates HTML for each request.
- Browser gets fully-rendered HTML.
- Example: Next.js with SSR

CSR VS SSR

Pros:

- Fast initial load (HTML ready)
- Better SEO
- Works without JS

Cons:

- Higher server load
- Can be slower for client interactions
- More complex setup

RESOURCES

- react hook form in deep :

https://youtube.com/playlist?list=PL03g4H_exuTpp0gtY-45oWvN79rvJIKzf&si=a4eoU5z6AA9YfZuE

- <https://blog.bitsrc.io/automatic-batching-in-react-18-what-you-should-know-d50141dc096e>

- <https://namansaxena-official.medium.com/react-virtual-dom-reconciliation-and-fiber-reconciler-cd33ceb0478e>

- <https://the-guild.dev/blog/react-hooks-system>

- <https://github.com/pomber/didact>

- https://youtube.com/playlist?list=PLC3y8-rFHvwi1AXijGTKM0BKtHzVC-LSK&si=Y3F36f_d40IxxxyNG

- <https://knock.app/blog/the-top-notification-libraries-for-react>

RESOURCES

- <https://gist.github.com/romain-trotard/76313af8170809970daa7ff9d87b0dd5>
- <https://www.youtube.com/watch?v=0ympFIwQFJw>
- <https://github.com/mithi/react-philosophies>
- <https://react-hooks-cheatsheet.com/>
- <https://www.freecodecamp.org/news/how-to-add-localization-to-your-react-app/>
- https://youtube.com/playlist?list=PLf16UKl7nR5AajcrY0e1niifJSAl_s3spDk&si=ueV-ejDZXNoMwBdo
- <https://youtu.be/cgwLP8y6cA8?si=MGS0Wsf14xfWdmKp>

LAB



1-Create a custom React hook named useFetch that fetches data from an API , The hook should manage three states: data, loading, and error

2-try to implement localization(en , ar)

THANK YOU